

## SEMESTER-V

| UMT503: AUTOMATIC CONTROL SYSTEMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |     |
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| <b>Course Objective:</b> To understand concepts of the mathematical modeling, feedback control, stability and analysis in time and frequency domains.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |   |   |     |
| <b>Syllabus</b><br><b>Basic Concepts:</b> Open and closed loop control systems, dynamics of mechanical systems and electromechanical systems (motors, gears), electric circuit models, heat and fluid-flow models, concept of transfer function, block diagrams and signal flow graphs.<br><b>Control hardware and their models:</b> potentiometers, synchros, LVDT, dc and ac servomotors, tacho-generators, and stepper motors.<br><b>Analysis:</b> Transient analysis of second order systems, Time domain specifications, Steady-state errors and error constants, Concepts of P, PD, PI and PID types of control and their realizations.<br><b>Stability:</b> Definition, Routh-Hurwitz criterion, Root locus techniques, Nyquist criterion, Bode plots, Relative stability, concepts of gain and phase margins.<br><b>Feedback Analysis:</b> The basic equations of control-Stability, Tracking, Regulation and Sensitivity, Control of steady-state errors for polynomial inputs, system type for tracking, regulation and disturbance rejection.<br><b>Compensation:</b> Lead, Lag and lag-lead compensators, Design of compensating networks for specified control system performance.<br><b>State Space Analysis:</b> Concepts of state, State variables and state models, State space equations, transfer function, Transfer model, State space representation of dynamic systems, State transition matrix, Decomposition of transfer function, Controllability and observability. |   |   |   |     |
| <b>Laboratory Work</b><br>Linear system simulator, Compensation design, D.C. position control and speed control, Synchro characteristics, Servo demonstration, Stepper motor, Potentiometer error detector, Rate control system, Series control system, Temperature control system, simulation examples of control problems with MATLAB/Simulink software.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |   |     |
| <b>Course Learning Outcomes (CLOs)</b><br>The students will be able to: <ol style="list-style-type: none"> <li>1. develop the mathematical model of the physical systems.</li> <li>2. analyse the response of the closed and open loop systems.</li> <li>3. analyse the stability of the closed and open loop systems.</li> <li>4. design the various kinds of compensator.</li> <li>5. develop and analyse state space models</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |   |   |     |
| <b>Text Books</b> <ol style="list-style-type: none"> <li>1. Nagrath, I.J. and Gopal, M., Control System Engineering, New Age International (P) Limited, Publishers (2003).</li> <li>2. Kuo, B.C., Automatic Control System, Prentice–Hall of India Private Limited (2002).</li> <li>3. Ogata, K., Modern Control Engineering, Prentice–Hall of India Private Limited (2001).</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |   |     |

*Approved in 109<sup>th</sup> meeting of the Senate held on March 16, 2023. Revised in 112<sup>th</sup> and 113<sup>th</sup> meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1<sup>st</sup> meeting of Academic Council held on September 11, 2025.*

**Reference Books**

1. Sinha, N.K., Control System, New Age International (P) Limited, Publishers (2002).
2. Franklin, G. F., Powell, J. D., Emami-Naeini, A., & Sanjay, H. S. Feedback control of dynamic systems. London: Pearson (2015).

**Evaluation Scheme:**

| S.No. | Evaluation Elements                              | Weightage (%) |
|-------|--------------------------------------------------|---------------|
| 1     | MST                                              | 25            |
| 2     | EST                                              | 40            |
| 3     | Sessional (Assignments//Quizzes/Lab Evaluations) | 35            |

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